

The Abstraction Gap Between Mathematics and Physics: The Hidden Compromise in Dimensional Analysis Resolved

Fredrik Bierich 
`fredrik.bierich@gmail.com`

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Circular Sector

$$\begin{aligned}
 s &= \lim_{n \rightarrow \infty} 2^n s_n \\
 s_n &= \sqrt{r^2 + r^2 - 2ra_n} \\
 a_n &= \frac{r\sqrt{r^2 - a_{n-1}^2}}{\sqrt{r^2 + r^2 - 2ra_{n-1}}} \\
 a_0 &= x \quad \text{for } -r \leq x \leq r
 \end{aligned}$$

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$$\begin{aligned}
 s &= \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \\
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 a_0 &= x \quad \text{for } -r \leq x \leq r
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 s &= \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \\
 a_n &= r \sqrt{\frac{(r + a_{n-1})(r - a_{n-1})}{2r(r - a_{n-1})}} \\
 a_0 &= x \quad \text{for } -r \leq x \leq r
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 s &= \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \\
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 \end{aligned}$$

$$\begin{aligned}
 s &= \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \\
 a_n &= r \sqrt{\frac{2r(r + a_{n-1})}{(2r)^2}} \\
 a_0 &= x \quad \text{for } -r \leq x \leq r
 \end{aligned}$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n}$$

$$a_n = r \sqrt{\frac{2r^2 + 2ra_{n-1}}{(2r)^2}}$$

$$a_0 = x \quad \text{for } -r \leq x \leq r$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n}$$

$$a_n = r \sqrt{\frac{\left(\sqrt{2r^2 + 2ra_{n-1}}\right)^2}{(2r)^2}}$$

$$a_0 = x \quad \text{for } -r \leq x \leq r$$

$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n}$$

$$a_n = r \sqrt{\left(\frac{\sqrt{2r^2 + 2ra_{n-1}}}{2r}\right)^2}$$

$$a_0 = x \quad \text{for } -r \leq x \leq r$$

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$$s = \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n}$$

$$a_n = \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2}$$

$$a_0 = x \quad \text{for } -r \leq x \leq r$$

$$A_{ORS_{n-1}} = \frac{\overline{RS_{n-1}OX_n}}{2} = \frac{\overline{OS_{n-1}RX_{n-1}}}{2}$$

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$$\overline{RS_{n-1}OX_n} = \overline{OS_{n-1}RX_{n-1}}$$

$$\frac{\overline{RS_{n-1}OX_n}}{\overline{RS_{n-1}}} = \frac{\overline{OS_{n-1}RX_{n-1}}}{\overline{RS_{n-1}}}$$

$$\overline{OX_n} = \frac{\overline{OS_{n-1}RX_{n-1}}}{\overline{RS_{n-1}}}$$

$$\overline{OX_n} = \frac{\overline{OS_{n-1}} \sqrt{\overline{OR}^2 - \overline{OX_{n-1}}^2}}{\overline{RS_{n-1}}}$$

$$\overline{OX_n} = \frac{\overline{OS_{n-1}} \sqrt{\overline{OR}^2 - \overline{OX_{n-1}}^2}}{\sqrt{\overline{OR}^2 + \overline{OS_{n-1}}^2 - 2\overline{OS_{n-1}OX_{n-1}}}}$$

$$\overline{RS_n}^2 = \overline{RX_n}^2 + \overline{X_nS_n}^2$$

$$\overline{RS_n} = \sqrt{\overline{RX_n}^2 + \overline{X_nS_n}^2}$$

$$\overline{RS_n} = \sqrt{\overline{OR}^2 - \overline{OX_n}^2 + \overline{X_nS_n}^2}$$

$$\overline{RS_n} = \sqrt{\overline{OR}^2 - \overline{OX_n}^2 + (\overline{OS_n} - \overline{OX_n})^2}$$

$$\overline{RS_n} = \sqrt{\overline{OR}^2 - \overline{OX_n}^2 + \overline{OS_n}^2 - 2\overline{OS_nOX_n} + \overline{OX_n}^2}$$

$$\overline{RS_n} = \sqrt{\overline{OR}^2 + \overline{OS_n}^2 - 2\overline{OS_nOX_n}}$$

$$\overline{OX_n}^2 + \overline{RX_n}^2 = \overline{OR}^2$$

$$\overline{OX_n}^2 + \overline{RX_n}^2 - \overline{OX_n}^2 = \overline{OR}^2 - \overline{OX_n}^2$$

$$\overline{RX_n}^2 = \overline{OR}^2 - \overline{OX_n}^2$$

$$\overline{RX_n} = \sqrt{\overline{OR}^2 - \overline{OX_n}^2}$$

$$A = \frac{rs}{2}$$

$$A_0 = \lim_{n \rightarrow \infty} 2^{n-1} A_{ORS_n S_{n-1}}$$

$$A_0 = \lim_{n \rightarrow \infty} 2^{n-1} \frac{\overline{RS_{n-1} OS_n}}{2}$$

$$A_0 = \frac{\lim_{n \rightarrow \infty} 2^{n-1} \overline{RS_{n-1} OS_n}}{2}$$

$$A_0 = \frac{\lim_{n \rightarrow \infty} (2^{n-1} \overline{RS_{n-1}}) \overline{OS_n}}{2}$$

$$A_0 = \frac{\lim_{n \rightarrow \infty} \overline{OS_n} (2^{n-1} \overline{RS_{n-1}})}{2}$$

$$A_0 = \frac{\left(\lim_{n \rightarrow \infty} \overline{OS_n} \right) \left(\lim_{n \rightarrow \infty} 2^{n-1} \overline{RS_{n-1}} \right)}{2}$$

$$A_0 = \frac{\overline{OR} \lim_{n \rightarrow \infty} 2^{n-1} \overline{RS_{n-1}}}{2}$$

$$A_0 = \frac{\overline{OR} \text{arc } RS_0}{2}$$

$$\begin{aligned} \arccos: \{(r, x) \in \mathbb{R}^2 \mid r > 0, -r \leq x \leq r\} &\rightarrow \mathbb{R} \\ (r, x) &\mapsto s \end{aligned}$$

$$\begin{aligned} \arccos(r, x) &= \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \\ a_n &= \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2} \\ a_0 &= x \end{aligned}$$

$$\begin{aligned}\cos: \{(r, s) \in \mathbb{R}^2 \mid r > 0\} &\rightarrow \mathbb{R} \\ (r, s) &\mapsto x\end{aligned}$$

$$\begin{aligned}s &= \lim_{n \rightarrow \infty} 2^n \sqrt{2r^2 - 2ra_n} \\ a_n &= \frac{\sqrt{2r^2 + 2ra_{n-1}}}{2} \\ a_0 &= \cos(r, s)\end{aligned}$$

$$\begin{aligned}s &= \lim_{m \rightarrow \infty} 2^m \sqrt{2r^2 - 2ra_{m,m}} \\ a_{m,n} &= \frac{\sqrt{2r^2 + 2ra_{m,n-1}}}{2} \quad \text{for } n \leq m \\ a_{m,0} &= \cos(r, s)\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n-1} &= \frac{\sqrt{2r^2 + 2ra_{m,n}}}{2} \quad \text{for } n \leq m \\ 2^m \sqrt{2r^2 - 2ra_{m,0}} &= s\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ 2a_{m,n-1} &= 2 \frac{\sqrt{2r^2 + 2ra_{m,n}}}{2} \quad \text{for } n \leq m \\ 2^m \sqrt{2r^2 - 2ra_{m,0}} &= s\end{aligned}$$

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$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ (2a_{m,n-1})^2 &= \left(\sqrt{2r^2 + 2ra_{m,n}}\right)^2 \quad \text{for } n \leq m \\ 2^m \sqrt{2r^2 - 2ra_{m,0}} &= s\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ 2^2 a_{m,n-1}^2 &= \left(\sqrt{2r^2 + 2ra_{m,n}}\right)^2 \quad \text{for } n \leq m \\ 2^m \sqrt{2r^2 - 2ra_{m,0}} &= s\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ 4a_{m,n-1}^2 &= \left(\sqrt{2r^2 + 2ra_{m,n}}\right)^2 \quad \text{for } n \leq m \\ 2^m \sqrt{2r^2 - 2ra_{m,0}} &= s\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ 4a_{m,n-1}^2 &= 2r^2 + 2ra_{m,n} \quad \text{for } n \leq m \\ 2^m \sqrt{2r^2 - 2ra_{m,0}} &= s\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ 4a_{m,n-1}^2 - 2r^2 &= 2r^2 + 2ra_{m,n} - 2r^2 \quad \text{for } n \leq m \\ 2^m \sqrt{2r^2 - 2ra_{m,0}} &= s\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ 4a_{m,n-1}^2 - 2r^2 &= 2ra_{m,n} \quad \text{for } n \leq m \\ 2^m \sqrt{2r^2 - 2ra_{m,0}} &= s\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ \frac{4a_{m,n-1}^2 - 2r^2}{2r} &= \frac{2ra_{m,n}}{2r} \quad \text{for } n \leq m \\ 2^m \sqrt{2r^2 - 2ra_{m,0}} &= s\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ \frac{4a_{m,n-1}^2}{2r} - \frac{2r^2}{2r} &= \frac{2ra_{m,n}}{2r} \quad \text{for } n \leq m \\ 2^m \sqrt{2r^2 - 2ra_{m,0}} &= s\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ \frac{2a_{m,n-1}^2}{r} - \frac{2r^2}{2r} &= \frac{2ra_{m,n}}{2r} \quad \text{for } n \leq m \\ 2^m \sqrt{2r^2 - 2ra_{m,0}} &= s\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ \frac{2a_{m,n-1}^2}{r} - r &= \frac{2ra_{m,n}}{2r} \quad \text{for } n \leq m \\ 2^m \sqrt{2r^2 - 2ra_{m,0}} &= s\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ \frac{2a_{m,n-1}^2}{r} - r &= a_{m,n} \quad \text{for } n \leq m \\ 2^m \sqrt{2r^2 - 2ra_{m,0}} &= s\end{aligned}$$

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$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \\ \sqrt{2r^2 - 2ra_{m,0}} &= \frac{s}{2^m}\end{aligned}$$

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$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \\ 2r^2 - 2ra_{m,0} &= \left(\frac{s}{2^m}\right)^2\end{aligned}$$

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$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \\ 2r^2 - 2ra_{m,0} - 2r^2 &= \frac{s^2}{2^{2m}} - 2r^2\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \\ -2ra_{m,0} &= \frac{s^2}{2^{2m}} - 2r^2\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \\ \frac{-2ra_{m,0}}{-2r} &= \frac{\left(\frac{s^2}{2^{2m}}\right) - 2r^2}{-2r}\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \\ a_{m,0} &= \frac{\left(\frac{s^2}{2^{2m}}\right) - 2r^2}{-2r}\end{aligned}$$

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$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \\ a_{m,0} &= \frac{s^2}{2^{2m}(-2r)} - \frac{2r^2}{-2r}\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \\ a_{m,0} &= \frac{s^2}{-2^{2m+1}r} - \frac{2r^2}{-2r}\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \\ a_{m,0} &= \frac{s^2}{-2^{2m+1}r} + \frac{-2r^2}{-2r}\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \\ a_{m,0} &= \frac{s^2}{-2^{2m+1}r} + r\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \\ a_{m,0} &= r + \frac{s^2}{-2^{2m+1}r}\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \\ a_{m,0} &= r - \left(-\frac{s^2}{-2^{2m+1}r} \right)\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \\ a_{m,0} &= r - \left(\frac{-s^2}{-2^{2m+1}r} \right)\end{aligned}$$

$$\begin{aligned}\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\ a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \\ a_{m,0} &= r - \frac{s^2}{2^{2m+1}r}\end{aligned}$$

$$\begin{aligned}a_{m,n} &= \frac{2a_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \\ a_{m,0} &= r - \frac{s^2}{2^{2m+1}r}\end{aligned}$$

$$\begin{aligned}r\hat{a}_{m,n} &= \frac{2(r\hat{a}_{m,n-1})^2}{r} - r \quad \text{for } n \leq m \\ r\hat{a}_{m,0} &= r - \frac{s^2}{2^{2m+1}r}\end{aligned}$$

$$\begin{aligned}r\hat{a}_{m,n} &= \frac{2r^2\hat{a}_{m,n-1}^2}{r} - r \quad \text{for } n \leq m \\ r\hat{a}_{m,0} &= r - \frac{s^2}{2^{2m+1}r}\end{aligned}$$

$$\begin{aligned}r\hat{a}_{m,n} &= 2r\hat{a}_{m,n-1}^2 - r \quad \text{for } n \leq m \\ r\hat{a}_{m,0} &= r - \frac{s^2}{2^{2m+1}r}\end{aligned}$$

$$\begin{aligned}\frac{r\hat{a}_{m,n}}{r} &= \frac{2r\hat{a}_{m,n-1}^2 - r}{r} \quad \text{for } n \leq m \\ r\hat{a}_{m,0} &= r - \frac{s^2}{2^{2m+1}r}\end{aligned}$$

$$\begin{aligned}\hat{a}_{m,n} &= \frac{2r\hat{a}_{m,n-1}^2 - r}{r} \quad \text{for } n \leq m \\ r\hat{a}_{m,0} &= r - \frac{s^2}{2^{2m+1}r}\end{aligned}$$

$$\begin{aligned}\hat{a}_{m,n} &= \frac{2r\hat{a}_{m,n-1}^2}{r} - \frac{r}{r} \quad \text{for } n \leq m \\ r\hat{a}_{m,0} &= r - \frac{s^2}{2^{2m+1}r}\end{aligned}$$

$$\begin{aligned}\hat{a}_{m,n} &= 2\hat{a}_{m,n-1}^2 - \frac{r}{r} \quad \text{for } n \leq m \\ r\hat{a}_{m,0} &= r - \frac{s^2}{2^{2m+1}r}\end{aligned}$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1 \quad \text{for } n \leq m$$

$$r\hat{a}_{m,0} = r - \frac{s^2}{2^{2m+1}r}$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1 \quad \text{for } n \leq m$$

$$\frac{r\hat{a}_{m,0}}{r} = \frac{r - \left(\frac{s^2}{2^{2m+1}r}\right)}{r}$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1 \quad \text{for } n \leq m$$

$$\hat{a}_{m,0} = \frac{r - \left(\frac{s^2}{2^{2m+1}r}\right)}{r}$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1 \quad \text{for } n \leq m$$

$$\hat{a}_{m,0} = \frac{r}{r} - \frac{\left(\frac{s^2}{2^{2m+1}r}\right)}{r}$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1 \quad \text{for } n \leq m$$

$$\hat{a}_{m,0} = 1 - \frac{\left(\frac{s^2}{2^{2m+1}r}\right)}{r}$$

$$\hat{a}_{m,n} = 2\hat{a}_{m,n-1}^2 - 1 \quad \text{for } n \leq m$$

$$\hat{a}_{m,0} = 1 - \frac{s^2}{2^{2m+1}r^2}$$

$$\hat{a}_{m,n} = T_2^{\circ n}(\hat{a}_{m,0}) \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = T_2^{\circ n} \left(1 - \frac{s^2}{2^{2m+1}r^2} \right) \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = T_{2^n} \left(1 - \frac{s^2}{2^{2m+1}r^2} \right) \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{T_{2^n}^{(k)}(1)}{k!} \left(\left(1 - \frac{s^2}{2^{2m+1}r^2} \right) - 1 \right)^k \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{T_{2^n}^{(k)}(1)}{k!} \left(1 - \frac{s^2}{2^{2m+1}r^2} - 1 \right)^k \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{T_{2^n}^{(k)}(1)}{k!} \left(-\frac{s^2}{2^{2m+1}r^2} \right)^k \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{2^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \left(-\frac{s^2}{2^{2m+1}r^2} \right)^k \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{2^k \left(-\frac{s^2}{2^{2m+1}r^2} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(2 \left(-\frac{s^2}{2^{2m+1}r^2} \right) \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(-2 \frac{s^2}{2^{2m+1}r^2} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(-\frac{2s^2}{2^{2m+1}r^2} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(-\frac{s^2}{2^{2m}r^2} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(-\frac{s^2}{(2^2)^m r^2} \right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(-\frac{s^2}{4^m r^2}\right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s^2}{4^m r^2}\right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{\left(\frac{s^2}{r^2}\right)}{4^m}\right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{\left(\frac{s}{r}\right)^2}{4^m}\right)^k \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \frac{\left(\left(\frac{s}{r}\right)^2\right)^k}{(4^m)^k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \frac{\left(\frac{s}{r}\right)^{2k}}{(4^m)^k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(4^m)^k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{\left(\frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(4^m)^k}\right)}{(2k)!} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(4^m)^k (2k)!} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)! (4^m)^k} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)! (4^m)^k} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)! \prod_{j=0}^{k-1} 4^m} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \prod_{j=0}^{k-1} \frac{(2^n)^2 - j^2}{4^m} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \prod_{j=0}^{k-1} \frac{(2^n)^2 - j^2}{4^m} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} ((2^n)^2 - j^2)}{(2k)!} \prod_{j=0}^{k-1} \frac{4^n - j^2}{4^m} \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} \left(\frac{4^n}{4^m} - \frac{j^2}{4^m}\right)}{(2k)!} \prod_{j=0}^{k-1} \left(\frac{4^n}{4^m} - \frac{j^2}{4^m}\right) \quad \text{for } n \leq m$$

$$\hat{a}_{m,n} = \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m}\right)}{(2k)!} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m}\right) \quad \text{for } n \leq m$$

$$r\hat{a}_{m,n} = r \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m}\right)}{(2k)!} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m}\right) \quad \text{for } n \leq m$$

$$a_{m,n} = r \sum_{k=0}^{2^n} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m}\right)}{(2k)!} \prod_{j=0}^{k-1} \left(4^{n-m} - \frac{j^2}{4^m}\right) \quad \text{for } n \leq m$$

$$\begin{aligned}
\cos(r, s) &= \lim_{m \rightarrow \infty} a_{m,m} \\
\cos(r, s) &= \lim_{m \rightarrow \infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^{m-j} - \frac{j^2}{4^m}\right) \\
\cos(r, s) &= \lim_{m \rightarrow \infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(4^0 - \frac{j^2}{4^m}\right) \\
\cos(r, s) &= \lim_{m \rightarrow \infty} r \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(1 - \frac{j^2}{4^m}\right) \\
\cos(r, s) &= r \lim_{m \rightarrow \infty} \sum_{k=0}^{2^m} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \prod_{j=0}^{k-1} \left(1 - \frac{j^2}{4^m}\right) \\
\cos(r, s) &= r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial \cos(r, s)}{\partial s} &= \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \right)}{\partial s} \\
\frac{\partial \cos(r, s)}{\partial s} &= \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \frac{s^{2k}}{r^{2k}}}{(2k)!} \right)}{\partial s} \\
\frac{\partial \cos(r, s)}{\partial s} &= \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k s^{2k}}{r^{2k} (2k)!} \right)}{\partial s} \\
\frac{\partial \cos(r, s)}{\partial s} &= \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k r s^{2k}}{r^{2k} (2k)!} \right)}{\partial s} \\
\frac{\partial \cos(r, s)}{\partial s} &= \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k s^{2k}}{r^{2k-1} (2k)!} \right)}{\partial s} \\
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=0}^{\infty} \frac{(-1)^k \frac{\partial s^{2k}}{\partial s}}{r^{2k-1} (2k)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=1}^{\infty} \frac{(-1)^k (2k) s^{2k-1}}{r^{2k-1} (2k)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \frac{s^{2k-1}}{r^{2k-1}}}{(2k)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \left(\frac{s}{r}\right)^{2k-1}}{(2k)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=1}^{\infty} \frac{(-1)^k (2k) \left(\frac{s}{r}\right)^{2k-1}}{(2k)(2k-1)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=1}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= \sum_{k=1}^{\infty} \frac{-(-1)^{k-1} \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= - \sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2k-1}}{(2k-1)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= - \sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2(k-1)+1}}{(2k-1)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= - \sum_{k=1}^{\infty} \frac{(-1)^{k-1} \left(\frac{s}{r}\right)^{2(k-1)+1}}{(2(k-1)+1)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= - \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \\
\frac{\partial \cos(r, s)}{\partial s} &= \frac{-r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!}}{r} \\
\frac{\partial \cos(r, s)}{\partial s} &= \frac{-\sin(r, s)}{r}
\end{aligned}$$

$$\begin{aligned} \arcsin: \{(r, y) \in \mathbb{R}^2 \mid r > 0, -r \leq y \leq r\} &\rightarrow \mathbb{R} \\ (r, y) &\mapsto s \end{aligned}$$

$$\arcsin(r, y) = \frac{\arcsin(r, y) + \arcsin(r, y)}{2}$$

$$\arcsin(r, y) = \frac{\arcsin(r, y) + \arcsin(r, y) + \arccos(r, y) - \arccos(r, y)}{2}$$

$$\arcsin(r, y) = \frac{\arcsin(r, y) + \arcsin(r, -y) + \arccos(r, -y) - \arccos(r, y)}{2}$$

$$\arcsin(r, y) = \frac{\arcsin(r, y) - \arcsin(r, y) + \arccos(r, -y) - \arccos(r, y)}{2}$$

$$\arcsin(r, y) = \frac{\arccos(r, -y) - \arccos(r, y)}{2}$$

$$\begin{aligned}\sin: \{(r, s) \in \mathbb{R}^2 \mid r > 0\} &\rightarrow \mathbb{R} \\ (r, s) &\mapsto y\end{aligned}$$

$$\begin{aligned}\cos(r, \alpha - \beta) &= a_{0,0} \\ a_{0,0} &= r - \frac{s^2}{2^{2 \cdot 0 + 1} r} \\ s &= \sqrt{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2}\end{aligned}$$

$$\begin{aligned}\cos(r, \alpha - \beta) &= a_{0,0} \\ a_{0,0} &= r - \frac{s^2}{2^{0+1} r} \\ s &= \sqrt{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2}\end{aligned}$$

$$\begin{aligned}\cos(r, \alpha - \beta) &= a_{0,0} \\ a_{0,0} &= r - \frac{s^2}{2r} \\ s &= \sqrt{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2}\end{aligned}$$

$$\begin{aligned}\cos(r, \alpha - \beta) &= a_{0,0} \\ a_{0,0} &= r - \frac{\left(\sqrt{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2}\right)^2}{2r}\end{aligned}$$

$$\cos(r, \alpha - \beta) = r - \frac{\left(\sqrt{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2}\right)^2}{2r}$$

$$\cos(r, \alpha - \beta) = r - \frac{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2}{2r}$$

$$\cos(r, \alpha - \beta) = \frac{2r^2}{2r} - \frac{(\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2}{2r}$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - ((\cos(r, \alpha) - \cos(r, \beta))^2 + (\sin(r, \alpha) - \sin(r, \beta))^2)}{2r}$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - (\cos(r, \alpha) - \cos(r, \beta))^2 - (\sin(r, \alpha) - \sin(r, \beta))^2}{2r}$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - (\cos^2(r, \alpha) - 2\cos(r, \alpha)\cos(r, \beta) + \cos^2(r, \beta)) - (\sin(r, \alpha) - \sin(r, \beta))^2}{2r}$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - (\cos^2(r, \alpha) - 2\cos(r, \alpha)\cos(r, \beta) + \cos^2(r, \beta)) - (\sin^2(r, \alpha) - 2\sin(r, \alpha)\sin(r, \beta) + \sin^2(r, \beta))}{2r}$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - \cos^2(r, \alpha) + 2\cos(r, \alpha)\cos(r, \beta) - \cos^2(r, \beta) - (\sin^2(r, \alpha) - 2\sin(r, \alpha)\sin(r, \beta) + \sin^2(r, \beta))}{2r}$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - \cos^2(r, \alpha) + 2\cos(r, \alpha)\cos(r, \beta) - \cos^2(r, \beta) - \sin^2(r, \alpha) + 2\sin(r, \alpha)\sin(r, \beta) - \sin^2(r, \beta)}{2r}$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - \cos^2(r, \alpha) - \sin^2(r, \alpha) - \cos^2(r, \beta) - \sin^2(r, \beta) + 2\cos(r, \alpha)\cos(r, \beta) + 2\sin(r, \alpha)\sin(r, \beta)}{2r}$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - (\cos^2(r, \alpha) + \sin^2(r, \alpha)) - \cos^2(r, \beta) - \sin^2(r, \beta) + 2\cos(r, \alpha)\cos(r, \beta) + 2\sin(r, \alpha)\sin(r, \beta)}{2r}$$

$$\cos(r, \alpha - \beta) = \frac{2r^2 - r^2 - \cos^2(r, \beta) - \sin^2(r, \beta) + 2\cos(r, \alpha)\cos(r, \beta) + 2\sin(r, \alpha)\sin(r, \beta)}{2r}$$

$$\cos(r, \alpha - \beta) = \frac{r^2 - \cos^2(r, \beta) - \sin^2(r, \beta) + 2\cos(r, \alpha)\cos(r, \beta) + 2\sin(r, \alpha)\sin(r, \beta)}{2r}$$

$$\cos(r, \alpha - \beta) = \frac{r^2 - (\cos^2(r, \beta) + \sin^2(r, \beta)) + 2\cos(r, \alpha)\cos(r, \beta) + 2\sin(r, \alpha)\sin(r, \beta)}{2r}$$

$$\cos(r, \alpha - \beta) = \frac{r^2 - r^2 + 2\cos(r, \alpha)\cos(r, \beta) + 2\sin(r, \alpha)\sin(r, \beta)}{2r}$$

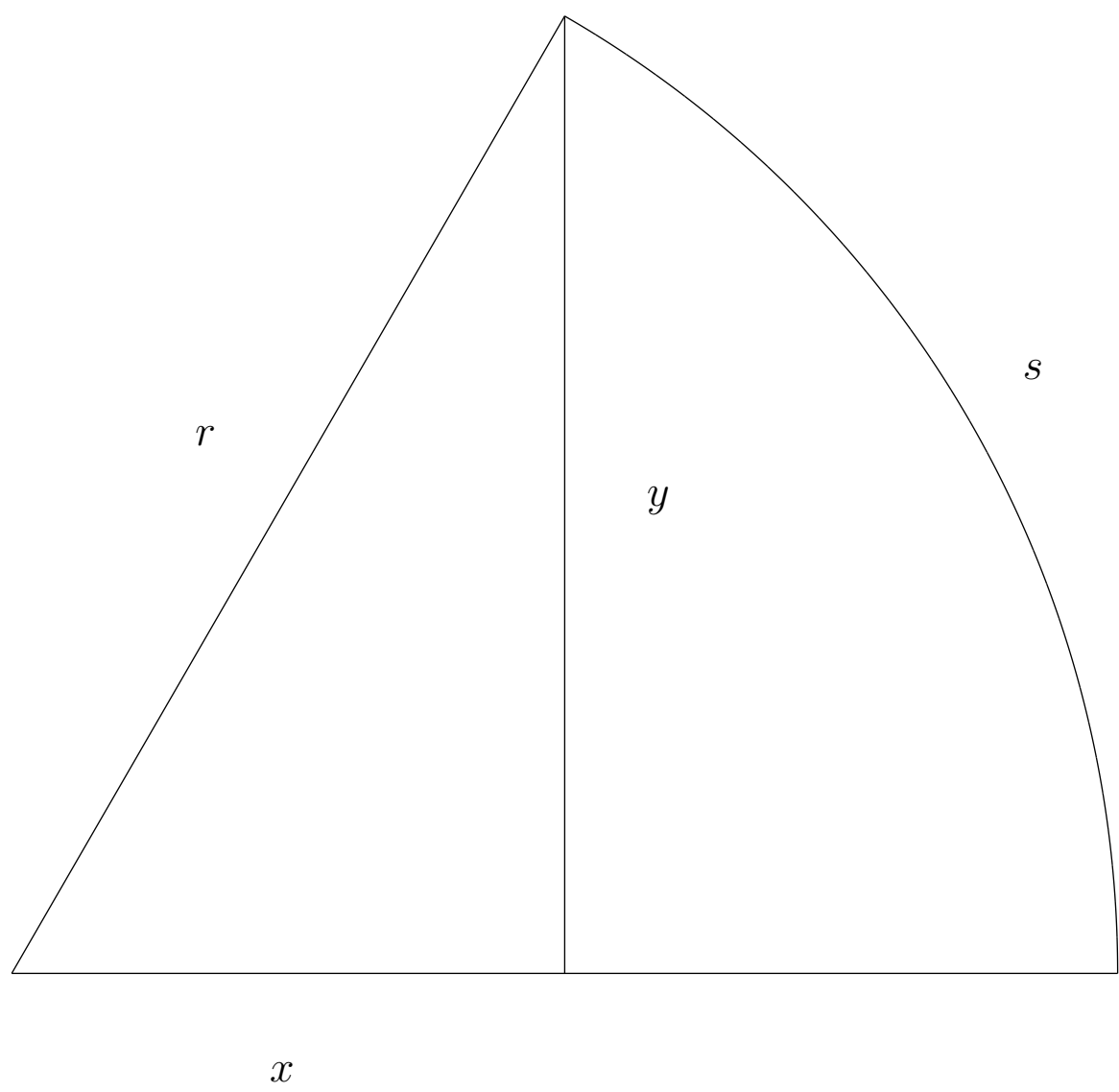
$$\cos(r, \alpha - \beta) = \frac{2\cos(r, \alpha)\cos(r, \beta) + 2\sin(r, \alpha)\sin(r, \beta)}{2r}$$

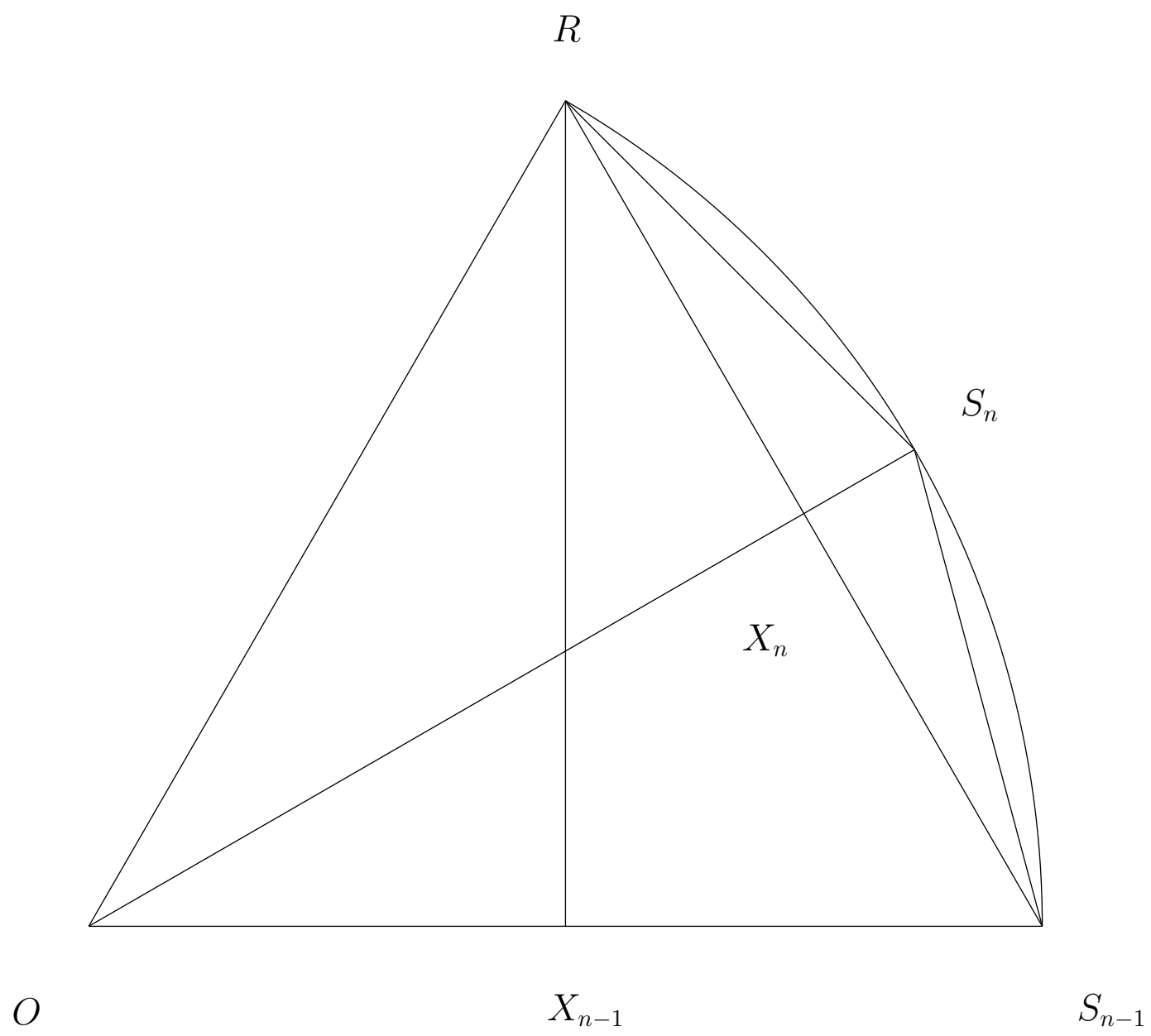
$$\cos(r, \alpha - \beta) = \frac{2(\cos(r, \alpha)\cos(r, \beta) + \sin(r, \alpha)\sin(r, \beta))}{2r}$$

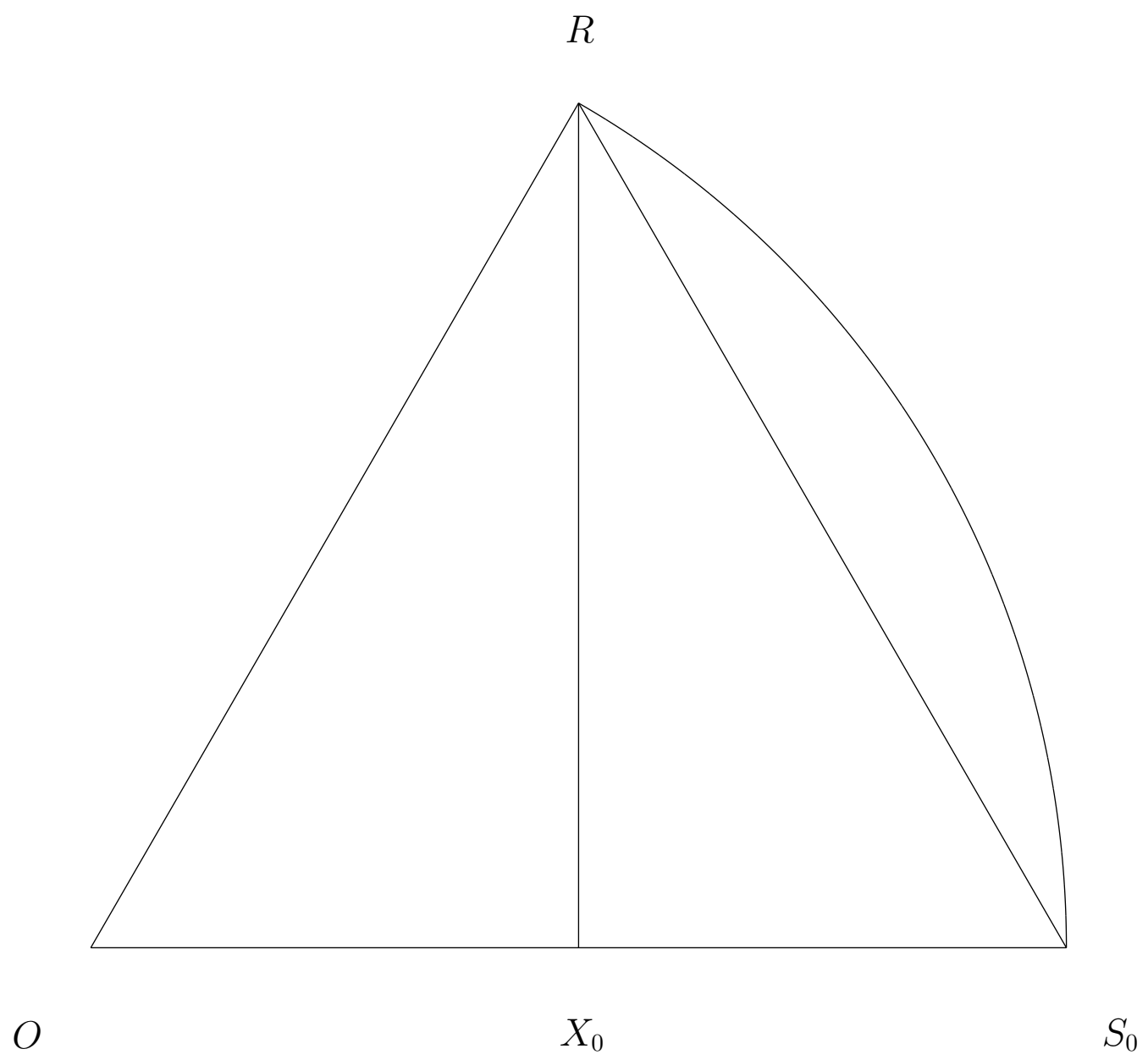
$$\cos(r, \alpha - \beta) = \frac{\cos(r, \alpha)\cos(r, \beta) + \sin(r, \alpha)\sin(r, \beta)}{r}$$

[illegible]

$$\begin{aligned}
\frac{\partial \sin(r, s)}{\partial s} &= \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k+1}}{(2k+1)!} \right)}{\partial s} \\
\frac{\partial \sin(r, s)}{\partial s} &= \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k \frac{s^{2k+1}}{r^{2k+1}}}{(2k+1)!} \right)}{\partial s} \\
\frac{\partial \sin(r, s)}{\partial s} &= \frac{\partial \left(r \sum_{k=0}^{\infty} \frac{(-1)^k s^{2k+1}}{r^{2k+1} (2k+1)!} \right)}{\partial s} \\
\frac{\partial \sin(r, s)}{\partial s} &= \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k r s^{2k+1}}{r^{2k+1} (2k+1)!} \right)}{\partial s} \\
\frac{\partial \sin(r, s)}{\partial s} &= \frac{\partial \left(\sum_{k=0}^{\infty} \frac{(-1)^k s^{2k+1}}{r^{2k} (2k+1)!} \right)}{\partial s} \\
\frac{\partial \sin(r, s)}{\partial s} &= \sum_{k=0}^{\infty} \frac{(-1)^k \frac{\partial s^{2k+1}}{\partial s}}{r^{2k} (2k+1)!} \\
\frac{\partial \sin(r, s)}{\partial s} &= \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) s^{2k}}{r^{2k} (2k+1)!} \\
\frac{\partial \sin(r, s)}{\partial s} &= \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \frac{s^{2k}}{r^{2k}}}{(2k+1)!} \\
\frac{\partial \sin(r, s)}{\partial s} &= \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \left(\frac{s}{r}\right)^{2k}}{(2k+1)!} \\
\frac{\partial \sin(r, s)}{\partial s} &= \sum_{k=0}^{\infty} \frac{(-1)^k (2k+1) \left(\frac{s}{r}\right)^{2k}}{(2k+1)(2k)!} \\
\frac{\partial \sin(r, s)}{\partial s} &= \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!} \\
\frac{\partial \sin(r, s)}{\partial s} &= \frac{r \sum_{k=0}^{\infty} \frac{(-1)^k \left(\frac{s}{r}\right)^{2k}}{(2k)!}}{r} \\
\frac{\partial \sin(r, s)}{\partial s} &= \frac{\cos(r, s)}{r}
\end{aligned}$$







Simple Harmonic Motion

$$[m] = \text{kg}$$

$$[\omega] = \text{m/s}$$

$$[k] = \text{N/m}$$

$$[r] = \text{m}$$

$$F = ma = -kx$$

$$ma = -kx$$

$$x(t) = \cos(r, \omega t)$$

$$a(t) = \frac{d^2 x}{dt^2}$$

$$a(t) = \frac{d^2 \cos(r, \omega t)}{dt^2}$$

$$a(t) = \frac{-\omega^2 \cos(r, \omega t)}{r^2}$$

$$a(x) = \frac{-\omega^2 x}{r^2}$$

$$m \frac{-\omega^2 x}{r^2} = -kx$$

$$\frac{m(-\omega^2 x)}{r^2} = -kx$$

$$\frac{-m\omega^2 x}{r^2} = -kx$$

$$\frac{-m\omega^2 x}{r^2} = \frac{-kr^2 x}{r^2}$$

$$m\omega^2 = kr^2$$

$$v(t) = \frac{dx}{dt}$$

$$v(t) = \frac{d \cos(r, \omega t)}{dt}$$

$$v(t) = \frac{-\omega \sin(r, \omega t)}{r}$$

$$K(t) = \frac{mv^2(t)}{2}$$

$$K(t) = \frac{m \left(\frac{-\omega \sin(r, \omega t)}{r} \right)^2}{2}$$

$$K(t) = \frac{m \frac{(-\omega \sin(r, \omega t))^2}{r^2}}{2}$$

$$K(t) = \frac{m \frac{(-\omega)^2 \sin^2(r, \omega t)}{r^2}}{2}$$

$$K(t) = \frac{m \frac{\omega^2 \sin^2(r, \omega t)}{r^2}}{2}$$

$$K(t) = \frac{\frac{m\omega^2 \sin^2(r, \omega t)}{r^2}}{2}$$

$$K(t) = \frac{\frac{kr^2 \sin^2(r, \omega t)}{r^2}}{2}$$

$$K(t) = \frac{k \sin^2(r, \omega t)}{2}$$

$$U(t) = \frac{kx^2(t)}{2}$$

$$U(t) = \frac{k \cos^2(r, \omega t)}{2}$$

$$E = K + U$$

$$E = \frac{k \sin^2(r, \omega t)}{2} + \frac{k \cos^2(r, \omega t)}{2}$$

$$E = \frac{k \sin^2(r, \omega t) + k \cos^2(r, \omega t)}{2}$$

$$E = \frac{k(\sin^2(r, \omega t) + \cos^2(r, \omega t))}{2}$$

$$E = \frac{kr^2}{2}$$

$$E = \frac{m\omega^2}{2}$$

Uniform Circular Motion

$$[m] = \text{kg}$$

$$[\omega] = \text{m/s}$$

$$[r] = \text{m}$$

$$s = \omega t$$

$$\omega = \frac{\Delta s}{\Delta t}$$

$$\vec{r} = (\vec{r}_x, \vec{r}_y)$$

$$\vec{r}(t) = (\cos(r, \omega t), \sin(r, \omega t))$$

$$\vec{v} = \frac{d\vec{r}}{dt}$$

$$\vec{v}(t) = \frac{d(\cos(r, \omega t), \sin(r, \omega t))}{dt}$$

$$\vec{v}(t) = \left(\frac{d \cos(r, \omega t)}{dt}, \frac{d \sin(r, \omega t)}{dt} \right)$$

$$\vec{v}(t) = \left(\frac{-\omega \sin(r, \omega t)}{r}, \frac{\omega \cos(r, \omega t)}{r} \right)$$

$$\vec{v}(t) = \frac{\omega (-\sin(r, \omega t), \cos(r, \omega t))}{r}$$

$$v = |\vec{v}|$$

$$v = \left| \frac{\omega (-\sin(r, \omega t), \cos(r, \omega t))}{r} \right|$$

$$v = \frac{|\omega (-\sin(r, \omega t), \cos(r, \omega t))|}{|r|}$$

$$v = \frac{|\omega| |(-\sin(r, \omega t), \cos(r, \omega t))|}{|r|}$$

$$v = \frac{\omega |(-\sin(r, \omega t), \cos(r, \omega t))|}{|r|}$$

$$v = \frac{\omega \sqrt{(-\sin(r, \omega t))^2 + \cos^2(r, \omega t)}}{|r|}$$

$$v = \frac{\omega \sqrt{\sin^2(r, \omega t) + \cos^2(r, \omega t)}}{|r|}$$

$$v = \frac{\omega \sqrt{r^2}}{|r|}$$

$$v = \frac{\omega |r|}{|r|}$$

$$v = \omega$$

$$E = \frac{mv^2}{2}$$

$$E = \frac{m\omega^2}{2}$$

$$\vec{a} = \frac{d^2\vec{r}}{dt^2}$$

$$\vec{a}(t) = \frac{d^2(\cos(r, \omega t), \sin(r, \omega t))}{dt^2}$$

$$\vec{a}(t) = \left(\frac{d^2 \cos(r, \omega t)}{dt^2}, \frac{d^2 \sin(r, \omega t)}{dt^2} \right)$$

$$\vec{a}(t) = \left(\frac{-\omega^2 \cos(r, \omega t)}{r^2}, \frac{-\omega^2 \sin(r, \omega t)}{r^2} \right)$$

$$\vec{a}(t) = \frac{-\omega^2 (\cos(r, \omega t), \sin(r, \omega t))}{r^2}$$

$$\vec{a} = \frac{-\omega^2 \vec{r}}{r^2}$$

$$a = |\vec{a}|$$

$$a = \left| \frac{-\omega^2 \vec{r}}{r^2} \right|$$

$$a = \frac{|-\omega^2 \vec{r}|}{|r^2|}$$

$$a = \frac{|-\omega^2| |\vec{r}|}{|r^2|}$$

$$a = \frac{\omega^2 |\vec{r}|}{|r^2|}$$

$$a = \frac{\omega^2 r}{|r^2|}$$

$$a = \frac{\omega^2 r}{r^2}$$

$$a = \frac{\omega^2}{r}$$

$$F = ma$$

$$F = m \frac{\omega^2}{r}$$

$$F = \frac{m\omega^2}{r}$$

